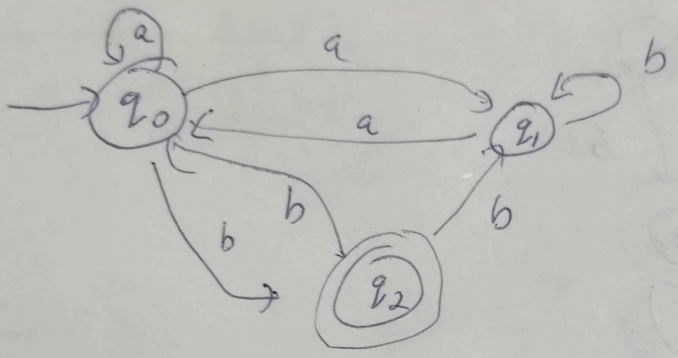
Assessment - Col-AT3

Q1) Consider the NFA given below which accepts some language L over the alphabet $\Sigma = (a, b)$. Construct a DFA that represents the same Language L.



Ans The NFA has states q_0, q_1, q_2 with q_0 as the initial state and q_2 as the final state

NFA Transition table

Current state	Input a	Input b
$\rightarrow q_0$	$\{q_0, q_1\}$	$\{q_2\}$
q_1	$\{q_0\}$	$\{q_1, q_2\}$
$*q_2$	$\emptyset$	$\{q_0, q_1\}$

Equivalent DFA transition table

DFA state	Input a	Input b
→ A = {q0}	B	C
B = {q0, q1}	B	D
*C = {q2}	E	B
*D = {q1, q2}	A	D
E = ∅	E	E

Here A, B, C, D and E sets of NFA states. DFA states containing q_2 are final states.

2) Let L be the Language of words of length atleast 2 whose last two letters are diffexence from each other. For example abaab is in L but not bb, Construct an NFA for L, Then draw the equivalent DFA.

Ans Given language L, consiting of strings of Length at least 2 over $\Sigma = \{a, b\}$ where the last two letters are different.

Examples:

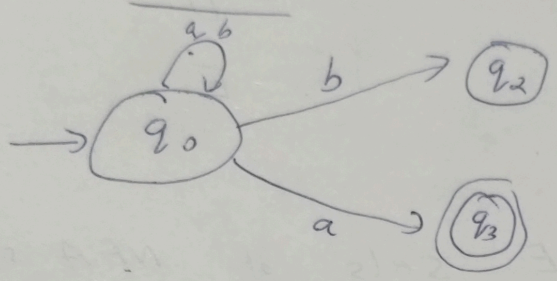
a a b - Accepted

a b - Accepted

b b - Rejected

a a - Rejected

NFA design



Formal definition

$$M = (Q, \Sigma, \delta, q_0, F)$$

$Q = \{q_0, q_1, q_2\}$

$\Sigma = \{a, b\}$

Initial state: q_0

Final states: $F = \{q_2\}$

NFA transition table

Current state	Input a	Input b
→ q_0	$\{q_0, q_1\}$	$\{q_0, q_2\}$
q_1	ϕ	$\{q_2\}$
* q_2	$\{q_1\}$	ϕ